

OFF GRID COMMUNICATION SYSTEM



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ABSTRACT

Most communication methods today depend on the internet. This can be a huge concern in locations like disaster-stricken areas, rural areas, and secure installations where the network infrastructure is either missing or not reliable. The purpose of this project is to create and build an Off-Grid Communication System that lets individuals talk to each other without utilizing the internet. Most modern ways to communicate rely on cloud infrastructure, which doesn't operate effectively when there is no internet. This means that in an emergency, communication is not very reliable. This project solves the problem by giving people away to talk to one another that is based on routers and operate totally offline. The system that was build indicates that it works and is valuable for keeping communication continuing in key situations, which makes it more reliable and easier to use.

The suggested system lets people send and receive text messages in real time, share multimedia like pictures and audio, and chat with groups. A mobile user interface makes it easier for people on the same network to talk to each other. An admin panel on a desktop computer in Ables you keep an eye on and control everything from one location, such as user permission, banning, and system supervision. The system makes an off-grid network by connecting long-range wireless routers to a linux-based local server. We used python for back-end development and PyQt for Admin Interface to acquire the features we needed. Flutter and Dart for mobile apps that let users interact.

We conducted both functional and integration testing on the messaging, multimedia, user and exploit management, and admin control functions. A Linux server on standard hardware and a mix of mobile devices connected by short-range wireless routers were used for validation. Results from the testing indicated that the product fulfilled its objectives of being able to transfer multimedia files, conduct real-time communication, and operate fully during offline and temporary connection scenarios. As for future product enhancements, even greater levels of network automation, provision of services based on the user's geographic location, services offered via the Cloud, when a network connection exists, would improve the product's application and effectiveness.

CERTIFICATE

Dated: _____

Final Approval

It is certified that project report 'Off grid communication system' submitted by **Muhammad Ahad Khan, Muhammad Zeeshan** for the partial fulfilment of the requirement of "**Bachelor's Degree in Computer Science**" is approved.

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DECLARATION

We hereby declare that our dissertation is entirely our work and genuine / original. We understand that in case of discovery of any PLAGIARISM at any stage, our group will be assigned an F (Fail) grade and it may result in withdrawal of our Bachelor's degree.

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Chapter 1

INTRODUCTION

Communication technology has become a huge part of our daily lives in the last few years. It makes it easier to talk to people, deal with emergencies, and work together. Most modern communication systems depend on being able to connect to the internet and employ cloud-based Infrastructure. These ways of talking to people fast and throughout the world area great, but they do not work well when there is no internet, the connection is spoity, or it is intentionally limited. These kinds of things happen a lot in places where calamities are common, like rural communities, warzones, and emergency operations. There are a lot of ways to talk to each other, but, many of them do not work well when you are not online, don't offer administrators enough power, or do not have modern features. This suggest that the current techniques of communicating are not working properly and that it's viral to find new ones.

The purpose of this project is to make an Off-Grid Communication system that does not need the to work. The suggested system has a Linux-based server and long-range wireless routers in the same region. This lets mobile devices talk to each other over a small local network. The system is in sync with new technologies for smart campuses, decentralized communication, and catastrophe management. The solution that was built has better features than current off-grid systems that work over the internet or peer-to-peer. These advantages include centralized administration, user management, multimedia communication, and a larger area of coverage. This makes it more helpful and dependable for real-world application that don-t need the internet.

Project Domain

This project is about computer networks and distributed systems, which are particularly significant for offline communication systems and networking technologies. This project is about how to solve challenges that happen when people need to talk to each other in places where isn't always internet access. The major purpose of the project is to communication system that uses wireless routers and a central server. The project wants to find a way to make offline saturation function by combining networking principles, client-server architecture, and distributed communication. The filed is related to both smart infrastructure technologies and emergency communication systems.

Problem Identification

Social networking sites and instant messaging application are two examples of modern communication system that need constant internet access and centralized cloud servers. When there is no internet or in an emergency, these technologies stop working, which means that users have no dependable way to talk to each other. Bridgefy and Briar are two examples of off-grid alternatives that mainly employ short-range technologies like Bluetooth. They don't have any administrative management or scalability. Most of the solutions that are available now also don't let you do advanced things like share multimedia, have group conversations, or manage people while they are aren't online. These restrictions make it very hard to communicate well at important times. To work without an internet connection, you need a dependable scalable, and feature-rich off-grid communication system.

Proposed Solution

The suggested answer is an Off-Grid Communication System that lets people talk to each other over a local network, even if they don't have internet connectivity. The systems is made up of a Linux-based server that talks to long-range wireless routers to make a local network for communication. People on the go can connect to the network over Wi-Fi, and a mobile app enables them chat to them other. Administrators can watch over and control the system from an admin panel on desktop. The server handles communication and user verification, storing all of the related data. This arrangement offers reliable communication, oversight, and coverage, which makes the system useful in remote and challenging locations.

Goals:

To design an offline option for group communication. To create a mobile application for real-time communication.

To enable image and audio message transfer. To allow group communication among users within a local area network.

To develop a control panel for the management and authorization of users. To watch a

system's function and reliability using wireless routers, and to improve remote communication.

Scope of the Project

This project focuses on designing a local communication system that integrates multimedia, group chat, administration oversight, and user control functionality, without an internet connection. The local communication system is built using long-range routers and a local server. The system is intended to support local area network communication that operates independently from public or global networks. This type of system is built for use in communications and the ability to transmit and receive information in real-time via a network without the use of the Internet in cases of disaster management, rural communications, military operations, and smart campus applications.

Effectiveness / Usefulness of the Systems

Compared to other systems, the Off-Grid Communication System supplies the benefits of guaranteed reliable communication in the absence of the internet. While others systems cease to function in the absence of the internet, this technology supports the permanent accessibility of on-site networks. In practice this technology gains value by supporting the group's ability to perform administrative actions, as well as their ability to partake in the group multimedia chat. Because reliable communication is continuously desired, this technology maintains highly beneficial value to emergent response teams, isolated communities, and the protective industry. This technology is helpful as it has the ability to operate across the internet on customized routers, and has the ability to cover a wide communication range.

Resource Requirement

This part explains the hardware and software that are needed to make and use the proposed system. These resources make sure the system runs smoothly, works reliably, and may grow as needed.

Hardware Requirement

For development and implementation, the system needs ordinary computer hardware. A Linux-based server is needed to handle routing and communication, and end consumers use mobile devices. Long-range wireless routers are used to set up the network that doesn't require the grid. Table 1.1 shows a summary of the hardware requirements.

Hardware Component	Specification
Server Machine	Core i5 / i7, 16GB RAM, 512GB SSD
Mobile Devices	Android smartphones
Routers	Long-range Wi-Fi routers (≈ 5 km range)
Network Interface	Wireless LAN adapters

Software Requirements

To build the system, you need a lot of different software tools and technologies. Python is utilized to build the backend and desktop applications, and mobile frontend technologies are employed to build the user interface. The server runs on Linux. Table 1.2 shows the software tools that are needed.

Software	Purpose
Linux (Ubuntu)	Server operating system
VS Code	Development environment
Android OS	Mobile client platform
PyQt	Desktop admin interface
Dart	Mobile application
Flutter	Mobile app framework

Data Requirements

There is no need for training or testing datasets for this project because it doesn't use machine learning techniques. The system processes real-time user-generated data, such as messages and multimedia content, on the local network.

Report Organization

This report is divided into various parts. Chapter 1 describes the project, its goals, what it will cover, and what resources it will need. Chapter 2 offers an extensive review of the existing literature on communication systems and related technologies. Chapter 3 describes the design and structure of the suggested solution's system. Chapter 4 goes into detail about how the system will be put into action. Chapter 5 is about testing, results, and how to measure performance. In the end, Chapter 6 cover up the report and is about possible changes.

Chapter 2
BACKGROUND AND EXISTING
SYSTEMS

In today's society, communication systems are incredibly crucial. However, most of the ones we have currently rely on the internet and central infrastructure. These tactics don't function in situations like natural disaster, rural areas, or safe zone. This chapter looks at existing research and technologies for off-grid and emergency communication and finds holes that the suggested solution would fill.

The structure of this chapter is as follows: Section 2.1 talks about important research literature on off-grid and emergency communication systems, as well as their problems. Section 2.2 looks at systems and applications that other researchers and groups have already made. Section 2.3 talks about the challenges that were found in the work that has already been done. Lastly, Section 2.4 sets the limits and range of the proposed answer based on the gaps in the study that have been found.

Related Literature Review

A. Rakhmadi, S. Wibowo, D. Kurniawan. [1] This project develops a prototype for an off-grid communication system that integrates smartphones with microcontroller-based radios and antennas to facilitate messaging in the absence of cellular connection. The prototype validates fundamental message transmission and illustrates the viability of smartphone and microcontroller integration for short-range, resilient connections. Restricted Bluetooth range: Bluetooth operates effectively only over small distances (10–20 meters), necessitating users to remain in proximity to the Arduino device. Each user is required to transport Arduino boards, RF modules, and antennas, rendering the solution unfeasible and expensive. Arduino and RF modules operate continuously, resulting in swift battery depletion. The complexity of message routing escalates with the growth of users, resulting in diminished performance. A central control mechanism for network management is absent. Our proposed approach eradicates reliance on supplementary hardware (Arduino, RF modules) and employs a solely smartphone-based MANET architecture. Instead of Bluetooth, Wi-fi Direct and device to device communication are used. These have a longer range and faster data rates. Also, energy efficient routing algorithms use less battery power. The system is built to handle a lot of users. A stream line monitoring system makes things more reliable.

R. Kumar, A. Sharma. [2] This android app uses Wi-Fi Direct to connect devices so that individuals can send emergency messages. Show that being able to quickly pair devices

and send messages is useful in situations that need to be handled right way. Wi-Fi Direct only works over short distances, usually 50-100 meters, so only devices that are close by can connect to it. Limitations of single group: Wi-Fi Direct need a group owner, and if that individual goes out of range, the network stop working. Issues with scaling: Wi-Fi Direct does not work with massive ad-hoc networks or multi-hop communication Energy use and complexity: This is a more advanced system and will use more batteries as it will continuously scan and manage links. An improved design will offer better range and connectivity through the use of optimized multi-hop routing and adaptive group management. Energy-efficient protocols increase battery life, while dynamic peer identification and routing address reliability concerns. Monitoring tools make network with a lot of traffic perform better.

J. Patel, L. Thompson, M. Rodriguez. [3] In the event of a disaster, a hybrid mesh that uses both Bluetooth and Wi-Fi could let people stay in touch when infrastructure is damaged. Demonstrates that hybrid mesh techniques enhance the strength and reliability of local connections. Not enough real-world testing. Relying on LoRa hardware. Not much support for app platforms. Low data and bandwidth speeds. Our system does not need any special hardware because it uses cell phones to talk to each other. It also has scalable multi-hop routing, uses less energy, and works with mobile apps on all platforms.

S. Khan, A. Mehmood. R. Iqbal. [4] This is about search and rescue communications that use Android and Wi-Fi Direct to let individuals' text one other even when they are not online. The technology indicates that responder in the area can reliably talk to each other and let each other know where they are in the short term. Limited range of communication: Wi-Fi Direct only works properly in a small area, therefor the signal does not reach far away areas. Issues with device compatibility: Not all smart phones support Wi-Fi Direct in the same way, which could make it hard to connect. The system only has basics single-hop Wi-Fi Direct connectivity and does not have good multi-hop mesh network functionality. Battery life: Talking on the phone and finding new devices all the time burns up a lot of battery life. Scalable Multi-Hop Communication: Our solutions combine multi-hop routing and mesh architecture to let devices talk to one other over longer distance and stay connected. Pure Smart phone system without extra Modules: Our system only works with smart phones and does not need any extra hardware, such additional radios. Energy efficient protocols: Our communication protocols use routing that saves energy. This makes batteries

last longer and make it easier to talk to each other in an emergency. The solution works on Android, among other platforms, and uses adaptive algorithms to make sure that devices can function together.

M. Ali, H. Saddique, S. Ahmed. [5] This is a test version of an Android app for managing disasters that uses Wi-Fi Direct and works effectively even when the connection is weak. Wi-Fi Direct is perfect for fast local communication and exchanges with little delay. Limited range of communication: Wi-Fi Direct can only connect to devices that are close by. Single-Hop Communication: The normal approach to use Wi-Fi Direct does not function well for multi-hop mesh networking, which makes the network less useful. Issues with compatibility: Wi-Fi Direct does not work the same way on all smart phones, which makes it challenging to connect and utilize. Using energy: Wi-Fi Direct requires a lot of battery power because it continues scanning and making groups. Multi-Hop Support: Our system has a better range and connections than Wi-Fi Direct since it uses optimize multi-hop mesh architecture. Scalable connectivity: Our design has dynamic routing and peer discovery, which makes sure that connection are strong in network with a lot of users. Energy Efficiency: The algorithms in our communication protocols save energy, which makes the battery last longer. Better compatibility with platforms: Our solution work with Android.

Y. Zhang, L. Chen, H. Wang. [6] This is about a temporary Wi-Fi network that let you test reliable connections between nodes and lets you rout via several hops. When there is a no infrastructure, multi-hop ad-hoc routing improve coverage, but it needs careful routing and resource management. Limited use case: The system is only for talking to sensors and equipment's; therefore, you cannot use it to talk to people in an emergency on your phone. The effective coverage range is limited about 50 meters in complicated indoor spaces, thus it is not practical for outside situation. The paper is largely on how to measure hardware, and it does not say much about what to do in an emergency for smart phones and people who use mobile apps. Restriction on transmission speed: The actual speed of transmissions are lower than the theoretical speeds because of restrictions in the design of hardware and software. Smartphone Centric Design: Our proposed system is mainly for smartphone users who need to communicate in an emergency without being connected to the internet. Better emergency range: Our method uses multi-hop mesh and optimize routing to make the range and network coverage better in case of emergency. Our answer is simple mobile app for

Android and does not need any extra hardware. Energy and Scalability: Our protocols are used communication methods that are both energy efficient and scalable. This keeps batteries from running out and network from growing too busy.

P. Jhonson, R. Smith, K. Brown. [7] A review of the literature on how ad-hoc networks are used in disasters, including, methodologies and typical challenges. This confirms that ad-hoc solutions are useful when infrastructure fails, but it also points out problems with scalability, power and management. Theoretical/Simulation-Based: The study is based only on data from reviews and simulations, and there has not been enough testing of mobile deployment in the real world. The design is built on broad ideas for ad-hoc networks and does not deal about smartphone apps for emergency communication. Security and QoS Challenges: The document leaves questions of safety and authentication and QoS in real time unanswered. The consequences of design and mobility of real world devices are faced. Practical multi-hop routing: Our design will use optimized multi-hop routing and mesh networking, allowing us to create a system with real-world, as opposed to just theoretical or simulated, network links. Safety and trust: Our designs will incorporate secure messaging and authentication methods that are absent from the paper. Energy and Quality of Service Improvement: Our system will provide powered Communication Quality of Service protocols and management.

S. Kumar, M. Abdul Rehman. [8] . It includes a review of the issues associated with multi-hop and ad hoc networking and routing techniques balancing trade-offs between scalability and performance. Describes a number of significant routing problems and offers solutions. Identifies the issues of practical design in significant detail. General review, practical issues: The paper is little more than a theoretical literature review that lacks any real-world applications, such as research based smartphone applications, etc. Lack of focus on emergency applications: The paper provides a general survey of different ad hoc paradigms and focuses little on the structure of emergency communication systems. No Performance Metrics: As a review study, this one does not provide any real performance data on metrics such as latency, throughput, and energy consumption. Security and Reliability: Coverage is inadequate for a number of emergency communication features specifically, reliability, and security and even more, authentication. Easy Smartphone Adoption: Our method offers users the ability to leap right to the emergency communication system. Most importantly, this allows them to use their mobile phones, even if there is no Internet connectivity. Mobile-centric Emergency Use Case: Our study is more granular, understanding how users of mobile phones for multiple communications first use their devices to build temporary networks to relay messages. Performance Metrics: Our study adequately captures relevant real-world

performance metrics, energy consumption, and reliability. Encrypted Messaging and Security: Our approach provides emergency communication of authenticated and encrypted messages.

J. Fisher, A. Gupta, M. Lee. [9] Hybrid systems use a combination of peer-to-peer messaging and radio broadcasting approaches to enhance coverage in disaster situations. Hybrid designs allow for reach and redundancy that enhance the stability of a system. Bluetooth Range and Connections: Bluetooth can only reach short distances and only allow for mesh connections of 8. These connections can only send data slowly. Other Bluetooth solutions have the same problems and some may even have worse solutions. Scalability and Maintenance: In large mesh networks it is difficult to route the communications, maintain the system, and manage node failures. Pure smartphone implementation: Our approach only requires a smartphone, for which the extra hardware is especially easy to integrate. This is the essential variable for rapid deployment. Better multi-hop networking: We have also improved multi-hop networking in systems based on mesh networking that do not incorporate backbone nodes. Our approaches provide borderless networking for adaptive range networking, Bluetooth, and Wi-Fi Direct. Our solutions also employ integrated approaches, for which the primary variable is energy thanks to our optimizations. For our systems, the communications remain optimal and offer the greatest trade-off for the most energy, even with many users.

M. V. Ramesh, A. Jacob, R. D. Arya Devi. [10] The goal of this study is to create an emergency communication model that uses Participatory Sensing over Mobile Ad Hoc Networks (MANETs) that empower mobile phones to collect and share data with each other when traditional infrastructures are compromised. The study shows that in the absence of cellular networks, participating in MANET based participatory sensing empowers mobile phones to send messages, create situational awareness, and send emergency alerts. The findings show being connected to the cellular network is not a prerequisite when important messages are sent - mobile devices can create a network and seamlessly share information. Mobile devices that are continually sensing and relaying are quickly draining mobile usage; this is however, the high cost of the MANET. Scalability issue:

The network gets slower as more people utilize it. Unstable connectivity: Moving nodes often breaks links. No centralized oversight: There isn't a good way to control and watch over the network. Our suggested solution uses communication mechanisms that use less energy and battery life. Also, improved routing methods are used to make the network more stable and scalable.

Table 2.1: Summary of Reviewed Literature

Year	Authors	Contribution	Techniques used	Limitations
2021	A. Rakhmadi, S. Wibowo, D. Kurniawan	Developed an off-grid communication system integrating smartphones with Arduino and RF modules to enable offline messaging during disasters.	Arduino UNO, Bluetooth (HC-05), RF transceiver (HCW69), microstrip antenna (915 MHz), Android (Java)	Dependency on extra hardware, limited Bluetooth range, high energy consumption, scalability issues, no centralized monitoring
2016	R. Kumar, A. Sharma	Designed a mobile application using Wi-Fi Direct for offline emergency communication and location sharing.	Wi-Fi Direct, peer-to-peer networking, Android mobile application	Short communication range, single-hop connectivity, lack of multi-hop routing, high battery consumption
2025	J. Patel, L. Thompson, M. Rodriguez	Proposed an off-grid emergency communication system using LoRa technology combined with smartphone applications.	LoRa radio communication, smartphone integration, mesh networking	Hardware dependency, low bandwidth, limited field testing, restricted multimedia support
2025	S. Khan, A. Mehmood, R. Iqbal	Proposed an Android-based offline	Wi-Fi Direct, Android application,	Limited communication range, device

		communication system using Wi-Fi Direct to support search and rescue operations.	GPS location sharing, offline voice communication	compatibility issues, no multi-hop routing, high battery consumption
2019	M. Ali, H. Siddiqui, S. Ahmed	Developed an offline disaster communication system enabling message and location sharing using Wi-Fi Direct.	Wi-Fi Direct, Android application, peer-to-peer networking	Short range, single-hop communication, device compatibility problems, high energy usage
2024	Y. Zhang, L. Chen, H. Wang	Proposed a wireless ad hoc network platform for multi-bus instruments enabling infrastructure-less multi-hop communication.	ESP-12H Wi-Fi module, ad hoc networking, tree topology mesh design	Limited application context, short indoor range, lack of smartphone focus, constrained transmission speed
2021	P. Johnson, R. Smith, K. Brown	Reviewed the use of ad hoc networks for disaster communication and analyzed performance metrics.	Ad hoc networking concepts, MANET analysis, performance evaluation metrics	Theoretical and simulation-based, no smartphone-specific implementation, limited security and QoS discussion
2015	S. Kumar, M. Abdel-Rahman	Provided a comprehensive survey of multihop ad hoc networks and their role in disaster response.	MANET, VANET, DTN, WSN (survey and comparative analysis)	No practical implementation, no performance experimentation, limited focus on security and reliability
2018/19	J. Fisher, A. Gupta, M. Lee	Proposed a hybrid offline messaging system combining peer-to-peer communication with radio	P2P networking, Bluetooth, Wi-Fi Direct, radio broadcast backbone	Hardware dependency, complex deployment, Bluetooth range limitations, scalability and maintenance challenges

		backbone nodes.		
2012	M. V. Ramesh, A. Jacob, R. D. Arya Devi	Proposed a participatory sensing-based emergency communication system using MANET to enable message exchange during disasters without cellular infrastructure.	MANET, participatory sensing, peer-to-peer communication, Wi-Fi/Bluetooth, Android application	High energy consumption, unstable connectivity due to mobility, scalability issues, lack of centralized monitoring and security mechanisms

Related Systems / Applications

System 1: Bridgefy

Bridgefy is a messaging app that works without the internet. It uses Bluetooth to let people talk to each other without needing to be connected to the internet. It lets people send text messages inside a limited area. Bridgefy makes it easy to talk to people who are close by, but it doesn't offer long-range networking, centralized oversight, or multimedia sharing.

System 2: Briar

Briar is a tool that helps journalist and activist talk to each other without anybody else knowing. When you are not connected to Wi-Fi or Bluetooth, it's easier to get online. Briar offers good security protections, but it does not have an administration panel, centralized user management, or many networking choices. The technique is hard to understand for people who are not tech/savvy.

Table 2.2: Summary of Existing System

Year	System	Contribution	Tool/Technologies	Limitations	Applications
2016	Bridgefy	Offline Messaging	Bluetooth	Short Range, No Admin control	Emergency Messaging
2018	Briar	Secure offline communication	Wi-Fi Direct, Bluetooth	Complex setup, no monitoring	Secure communication

Identified Problem from Existing work

The present system as of today and the research conducted have observed many bounds. Most current methods of communicating off the grid utilize short-range networks such as Bluetooth, which means communicating users cannot be positioned far apart. Many systems are incapable of user management, activity tracking, or centralized control. Multimedia communication, long-distance and group texting, and other forms of communication are often impossible. Large scale and ease of use remain a particular challenge. The present state of these systems highlights an evident research need for scalable, long-range, and robust off-grid communication systems featuring administrative control. This is the primary justification of the proposed framework.

Selected Boundary for Proposed Solution

This system aims to develop an off-grid communication solution for real-time texting, multimedia dissemination, and group discussions on a private network. A centralized Linux server with long-range wireless routers has been incorporated to help ease connection hurdles and provide a compact solution to distance and communication chaos problems. To resolve issues that are present in other competing systems, additional administrative capabilities of user authorization, system banning and monitoring, have been incorporated.

The suggested solution does not include certain features on purpose. Because of its complexity and lack of resources, the system doesn't have global internet access, public cloud integration, or better encryption. GPS tracking and AI-based analytics are not part of the present scope. These limits set a clear limit for the project while still allowing for possible future changes.

Chapter 3

SYSTEM REQUIREMENT AND SPECIFICATIONS

This chapter basically define the system requirements and specifications of our Off-Grid Communication System. This concerns the key system of our project modules, functional requirements, and non-functional requirements is success of the project. Our objective is to detail system modules and expected system actions in J scenarios.

This chapter represents the following: Section 3.1, which contains full system requirements, and Section 3.2 which show the overall view of the system modules.

Part 3.3 talks about the system's functional demands or software features. Section 3.4 lists the non-functional requirements that define the quality features of the system that was made.

3.1 System Specifications

System specifications spell out what the proposed system needs in terms of technology and how it will work. These requirements spell out the system's operational environment, the hardware and software it needs, and the constraints on its performance. The specifications for the system in this project include the type of network, the operating systems, the programming languages, the user interfaces, and the communication protocols. Clearly defining these criteria ensures that the system is built and put into place in a controlled and suitable setting.

3.2 System Modules

An autonomous unit that executes a specific job and works with other units to reach the system's overall goals is called a system module. Modular architecture makes the system easier to manage, more flexible, and more scalable. The intended Off-Grid Communication System is broken up into different modules to make it easier to understand who is responsible for what and to speed up development. Each module has a specific job to do, yet it can still work with other modules. Below are the different parts of the system.

3.2.1 Server Module

The Server Module is the most important part of the system. It takes care of user connections, message routing, sending multimedia data, and keeping track of system activity. This module works on a Linux server and lets mobile users talk to the administrative panel. It makes sure that messages are sent reliably across the local network, even if there is no internet access.

3.2.2 Mobile Client Module

The first Mobile Client Modules was for users to sign-up to send and receive messages. Users could also exchange various file types, participate in a group chat, and share and receive voice screencasts. This module connects to the local server via Wi-Fi routers, allowing for off-grid network and real-time communication.

3.2.3 Administration Panel Module

The Admin panel modules use of desktops to monitor and manage a system. This module gives communication a defined handle and offers safety. It makes the system more controlled and secure.

3.2.4 Network and Router Module

The Network and Router Module makes it possible for people to talk to each other without using the internet. It uses long-range wireless routers to establish a local network with better coverage. This module makes it possible for several devices to talk to each other at distances of many kilometers without needing to be connected to the internet.

3.3. Functional Requirement

Functional requirement tell us about the system and what should be able to do. These specs spell out what services, features, and behaviour of our system have. However, the functional requirements of our system is align with the project objectives and scope.

3.3.1 User Registration and Authentication

This system must check that users sign up or log in with a unique identity. The administrator will have the power to say yes or no to users request.

3.3.2 SMS Communication

The system allow the people on the same local network can send and also successfully received text messages in real time.

3.3.3 Multimedia Distribution

The system should let people send pictures, videos, audio messages, and data without being connected to the internet.

3.3.4 Communication in Groups

The technology should make it possible for group chat, which lets multiple users talk to each other at the same time in a single group.

3.3.5 Administrative Oversight and Regulation

This system has a communication system module which offers you the privilege to manage users, direct messages, and offers system insights.

3.4 Non-functional requirements

In this project Non-functional requirements tell us how our system performs an action, specify parameters such as ease of use, operational effectiveness, reliability, and safety, and set a benchmark. The criteria met fulfills the system's quality criteria.

3.4.1 Usability

HerUsability show that how simple a system is to use. The system is designed to have a simple access for mobile users as well as easy interfaces for the administrators. The extent of a new user's ease of, speed of, and self-reliant functional understanding of the system accounts for the determination of usability.

3.4.2 Performance

Performance refers to the system's capability for messaging, sending and receiving multimedia files. There is also a desire for the technology to create the capability of practically effortless real time synergy var the local network. The speed of the system's responsiveness and the time expended in the delivery of the message reflect the performance of this System.

3.4.3 Reliability

Here reliability is assure that, this system operates unhindered and correctly. The system is expected to maintain functionality even after the disruption of the internet and to keep communication among connected devices possible. Successful delivery of a message and the loss of the communication system are two of the many ways the system's reliability is measured.

3.4.4 Security

A measure of level of security of the system is also determined by the inability of unauthorized persons to access the system. For the purpose of safekeeping, the administrators invoked customized means for users to approve, to excommunicate and to supervise. The system's characteristic to inhibit un permissioned presence, is a measure of security.

3.4.5 Scalability

The purpose of scalability is check the capability of an specific system and their users and devices in a possibly unlimited dimension. The system is designed to accommodate many users, yet only to the limit of the router. The limit of the system is reached with the addition of users.

Chapter 4

SYSTEM MODELING AND DESIGN

In this chapter System design and analysis are important steps in making software systems because both are turn user needs into a clear and organized answer. The term system design is tell us about the overall structure of a system. System analysis, on the other hand, makes sure that all functional and non-functional needs are well understood and addressed. This step manages to clarify the system and provide a reliable course of action. Better models distinguish potential issues and uncertainties and address the need to rebuild and redevelop elements within the system. The system is an Off-Grid Communication System. This system is designed to support people in communicating without the need for internet and telephony networks. To support the communication of both text messages and multimedia, the system allows the creation of a local ad hoc network within users. The communication would be monitored and controlled during the process. In relation to the parts of this system, many design and consolidative analytical diagrams have been formulated to explain the relation of the system components, namely the Use Case diagrams, Activity diagrams, Data Flow Diagrams (DFD), System Sequence Diagrams, UML Sequence Diagrams, UML Class Diagrams, and UI Architectural Diagrams.

This chapter is structured as follows: 4.1 Summary of analysis and description about architecture of this system, and 4.2 the Use Case Diagrams and their significance and the rest of the sections contain additional modeling diagrams, including those that pertain to system operations, user and system interaction, architecture of this system.

4.1 System Design and Analysis

To address the need for clarity, completeness, and systematic description, it is necessary to present the design from various perspectives. This work will use various design and analytical models in order to illustrate the functionality, data flow, system structure, and system components.

The Use Case Diagram describes about the functional requirements of the project. Activity Diagram shows the system's internal processes, and Data Flow Diagrams show the understanding of data between processes and entities, System Sequence Diagrams and Sequence Diagrams describe the flow of messages and describe each module of the system. While the Class Diagram describes its static structure and atlast Architectural Diagram provides a better snapshot of its structure, and also the components, how they are deployed. All of these models provide the structure for the system.

4.2 Use Case Diagrams

Use Case Diagram show its functional requirements and describe how the

system communicates with the user. The diagram shows the system from a high level and therefore helps define the functional requirements. Use case diagrams provide a user's perspective of the system and describe the system's functionality to stakeholders.

The Off-Grid Communication System relies heavily on user interaction; therefore, the use case diagram is essential. The diagram is divided into Admin and User. The user registers or joins the network, sends and receives messages, shares various media, and collaborates with the group. As the system owner, the Admin manages user access, issues or removes restrictions, supervises, and reviews communication, distributes system updates, and keep an eye on linked devices.

Fig 4.1 depicts the Use Case Diagram, for the Off-Grid Communication System. It illustrates how all actors communicate with their system. It shows how decisions are made, how the system is controlled, and how the role of actors is distributed.

The system use case diagram, provides a top-down overview of our system.

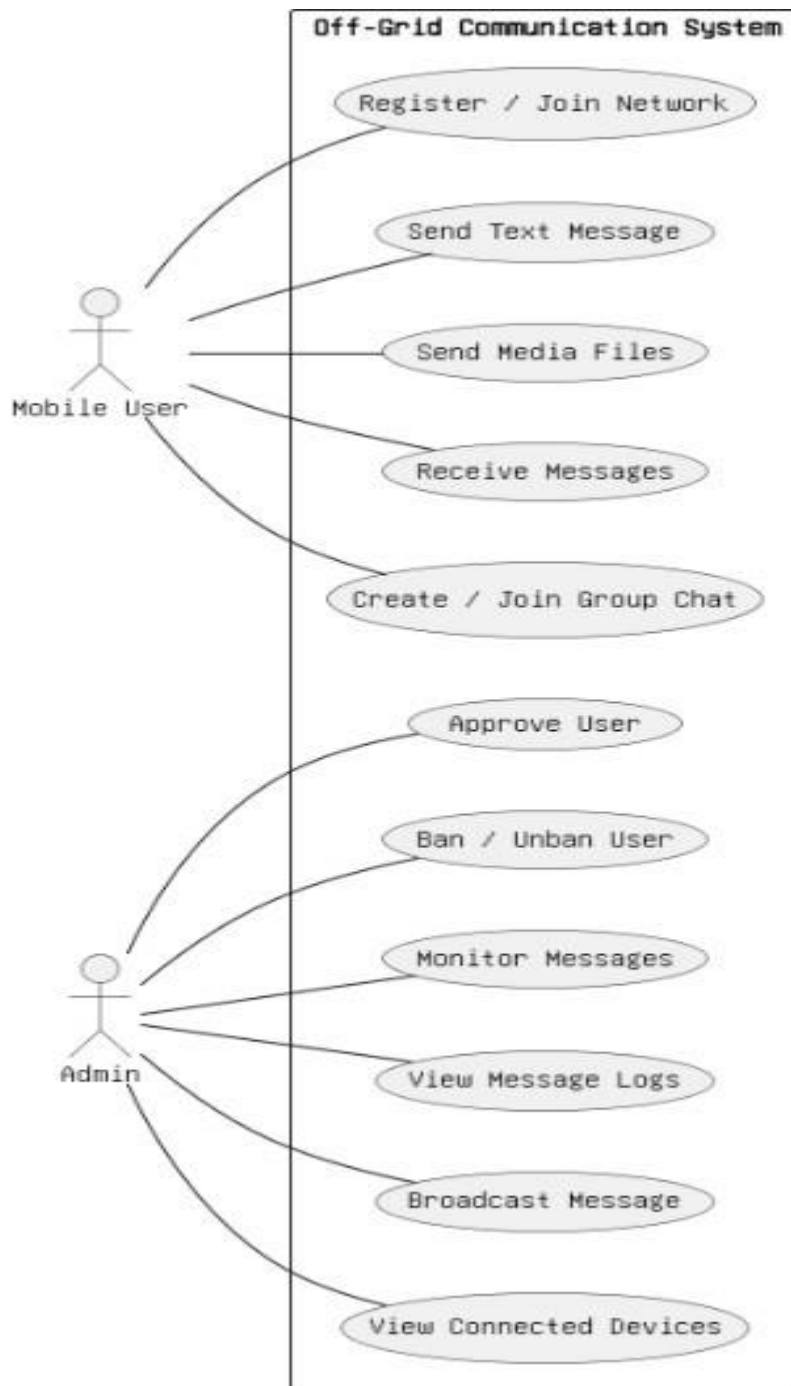


Figure 4.1: Main Use Case Diagram

4.3. Full Dress Use Case / Detailed Use Case

This use cases offer a clear and comprehensive view of the many systems and processes. They are used to understand the interactions of actors with a system, outlining the normal operations, exceptional cases, and system limitations. These use cases are particularly valuable for intricate activities that are multi- step processes and require several validations and decisions, as these cannot be simply represented in a use case diagram.

The proposed Off-Grid Communication System offers only a limited number of extensive use cases. Basic use cases, such as a "Login" use case, are not included as they are straightforward and require minimal

elaboration. The extensive use cases that were selected are Send Message, Send Media, Group Chat, and User Management (Ban/Unban). These use cases were selected because, unlike the simpler use cases, they require multiple actors, system validations, offline routing, and various forms of system administration, among others, and therefore require more extensive explanations and descriptions.

4.3.1 Send Message – Full Dress Use Case

The send message use case describes how a user sends a text message through the Off-Grid Communication System without internet connectivity. Details about this use case are described in Table 4.3.

Table 4.3 Full Dress Use Case – Send Message

Use Case Selection Comment	Description
Use Case Name	Send Message
Scope	Off-Grid Communication System
Level	User Goal
Primary Actor	User mobile Application
Stakeholders and Application	User: wants reliable message delivery without internet. Admin: wants controlled and logged communication
Pre conditions	User is connected to the local off-grid network and is approved by the admin.
Success Guarantee	Message is delivered to the intended recipient and logged on the server.
Main Success Scenario	1. User enter message text. 2. User can selects recipient. 3. System forwards message via routers. 4. Recipient receives message successfully.
Extensions	3a. Recipient offline → message queued. 3b. User banned → message rejected.
Special Requirements	Low latency, offline routing, message integrity.

4.3.2 Send Media (Image/Video/Voice) – Full Dress Use Case

Send Media use case explains the overall process of sending multimedia content over the off-grid network.

Table 4.4 Full Dress Use Case – Send Media

Use Case Selection Comment	Description
Use case name	Send Media
Scope	Off-Grid Communication
Level	User
Primary Actor	User
Stakeholders and Interests	Users need reliable file sharing; Admin monitors usage and storage.
Pre-conditions	User connected to off-grid network and storage available.
Success Guarantee	Media file successfully transferred and acknowledged.
Main Success Scenario	1. User can select media file. 2. System compresses data. 3. File transmitted through routers. 4. Recipient receives media
Extensions	2a. Large file → segmented transfer. 3a. Network interruption → resume transfer.
Special Requirements	File size handling, fault tolerance, data integrity.

4.3.3 Group Chat – Full Dress Use Case

This type of use case covers communication among multiple users in a group.

Table 4.5 Full Dress Use Case – Group Chat

Use Case Selection Comment	Description
Use case name	Group chat
Scope	Off-Grid Communication
Level	User Goal
Primary Actor	User
Stakeholders and Interests	Users want efficient group communication; Admin controls group access.
Pre-conditions	Group exists and user is approved member.
Success Guarantee	Message delivered to all group members.
Main Success Scenario	1. User choose group. 2. User sends message. 3. System broadcasts message locally
Extensions	3a. Member offline → message queue
Special Requirements	Efficient broadcasting, synchronization.

4.3.4 User Management (Ban/Unban) – Full Dress Use Case

In this use case explains administrative control over users.

Table 4.6 Full Dress Use Case – Ban/Unban User

Use Case Selection Comment	Description
Use Case Name	Ban/Unban User
Scope	Off-Grid Communication
Level	Sub-function
Primary Actor	Admin
Stakeholders and Interests	Admin ensures security; Users expect fair control
Pre-conditions	Admin logged into desktop system
Success Guarantee	User access updated successfully.
Main Success Scenario	1. Admin selects user. 2. Admin bans/unbans user. 3. System updates access rules.
Extensions	Invalid user → operation rejected.
Special Requirements	Access control, audit logging.

4.4 Activity Diagram

By the help of this diagram shows how activities flow through a system by showing the order of actions, decisions, and simultaneous operations. It gives a full understanding of system works from start to finish, making it easier to model business needs and system logic.

This project shows an activity diagram from the point of view of how users and systems talk to each other. It concentrates on offline messaging using a Linux server and multiple routers. The image depicts the entire process, beginning with the user and their mobile app, the server examining the message, the message being sent through intermediary routers, and the message finally being received by the recipient.

Figure 4.2 provides an activity diagram describing the system's processes. Once a user account is authenticated, a message or media is selected, and a delivery route through the off-grid network is established. The diagram also indicates points during the process where decisions, such as user authentication, or the presence of network resources, are required. The diagram provides an operational overview allowing users to confirm that all

components of the system are reliable and efficient.

Activity Diagram - Off-Grid Communication System



Figure 4.2: Activity Diagram

4.5 Data Flow Diagram (DFD)

A data flow diagram, or DFD, is basically show the graphical form of specific data flow in system. It also includes the sources, the destinations, placement of data, and the removal of data. Data flow diagrams are distinct from flow charts as they depict the movement of data and the transformation of data. Control flow or the making of decisions are irrelevant in DFDs.

Creating DFDs for this project is critical as they enable a grasp of how to manage user interactions, how to manage media and system logs for a system that operates offline. Given that the system operates offline using a Linux server and several routers, DFDs help visualize how data moves in the system.

For this exercise, we will perform Level 0 and Level 1 DFDs. Level 0 DFDs visualize the entire process as a single, while Level 1 DFDs represent decomposition of that process, in our case, authentication, message routing, media processing, and logging. Figures 4.3 and 4.4 illustrate how the system handles processing of the system's data.

4.5.1 Level 0 Data Flow Diagram

The Level 0 DFDs, or context diagrams, represent the Off-Grid Communication System as a single process. The typical system boundaries are illustrated by an external entity, which in our case is an Admin and Mobile Users, and the system itself. User data, which consists of messages, media, and authentication, are the inputs. User data, which consists of messages, media, and authentication, are the outputs.

DFD Level 0 - Off-Grid Communication System

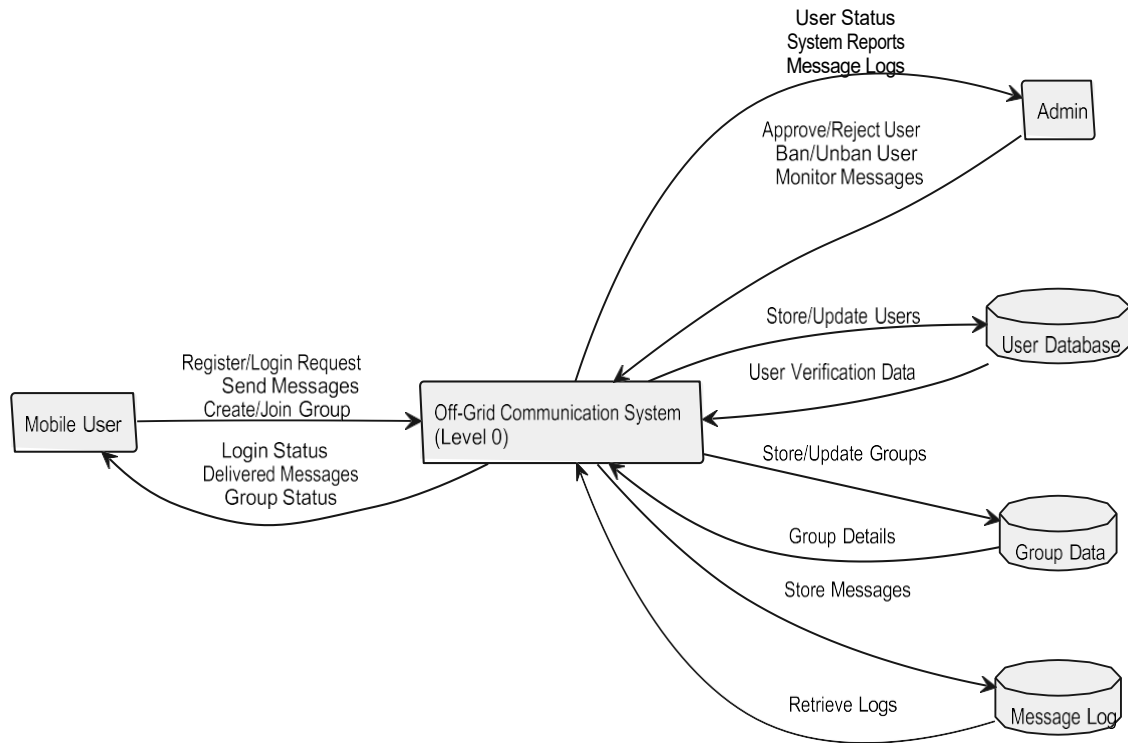


Figure 4.3: DFD Level 0

4.5.2 Level 1 Data Flow Diagram

The Level 1 DFD illustrates the main system decomposition into a complex process. The illustration presents the system subdivisions as User Authentication, Message and Media Processing, Routing through Routers, and Data Logging. Data Repositories within the system are denoted by the User Database and Message Logs, and the persistent nature of storage is illustrated by Message Logs.

DFD Level 1 - Off-Grid Communication System

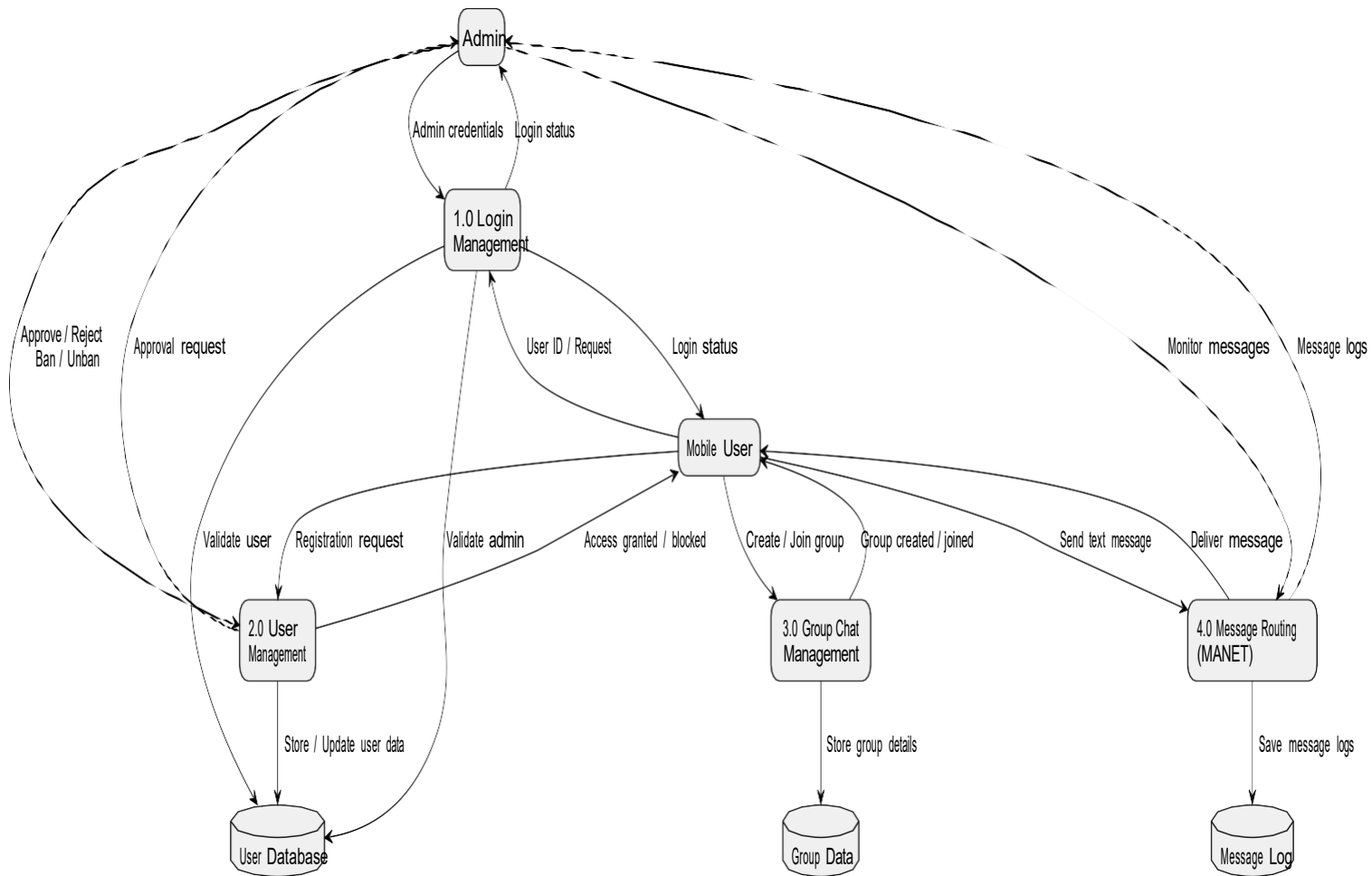


Figure 4.4: DFD Level 1

4.6 System Sequence Diagram

System Sequence Diagram (SSD) tell how external actors interact with the system in a series, focusing on input and output events. It shows how queries were sent to the system and how the system responded to them over time. SSDs are very useful for understanding how the system changes while it is running.

The goal of this project is to make SSDs that clearly show how messages are sent and received in the Off-Grid Communication System. The proposed system operates independently of internet connectivity, employing a Linux server and multiple routers for message transmission; hence, it is imperative to demonstrate the processing and routing of user requests within the system.

The System Sequence Diagram is based on all the use cases and shows how a message is sent from one mobile user to another through the Linux server and routers. The SSD shows how the user makes a request, how the system processes it, how it is sent over the network, and how it is finally delivered to the person who asked for it. Figure 4.5 shows the System Sequence Diagram showing how messages are sent and received in the system that was created.

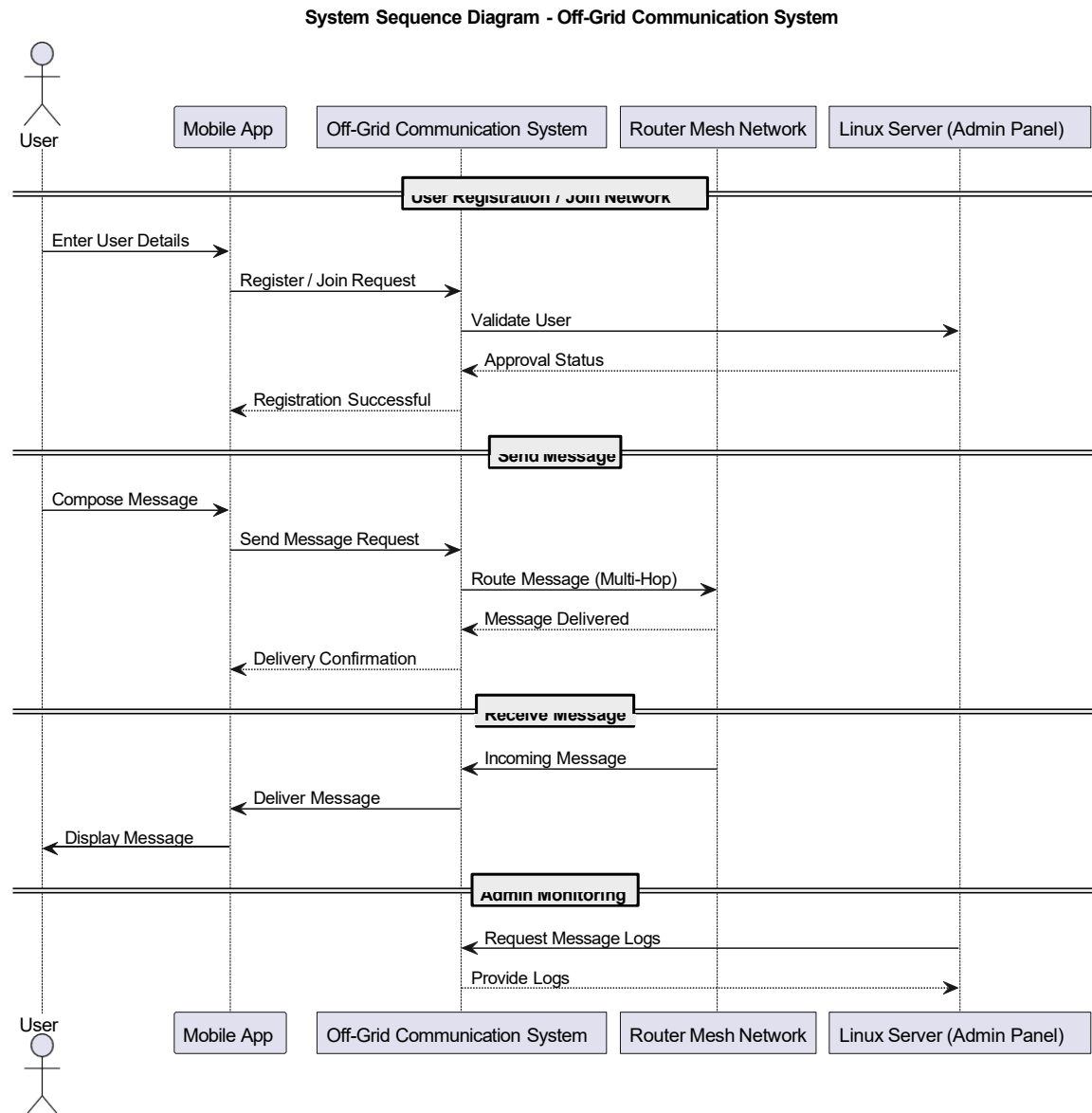


Figure 4.5: System Sequence Diagram

4.7 Sequence Diagram

A Sequence Diagram (SD) is a UML behaviour diagram and it shows how system parts interact with each other over time shows how external actors interact with the system as a black box, while a Sequence Diagram shows how internal objects and modules talk to one other to get a job done.

This project entails the development of a sequence diagram for the message and media transmission process within the Off-Grid Communication System. This situation was

chosen because it is important and complicated, involving many internal parts, such as the mobile app, Linux server, routing module, and logging service.

The sequence diagram shows how a mobile user sends a message request across several routers when offline, how it is logged for monitoring, and how it is finally transferred to the recipient. The figure also shows acknowledgements and status updates that were sent back to the person who started the process. Figure 4.6 shows the whole flow of contacts.

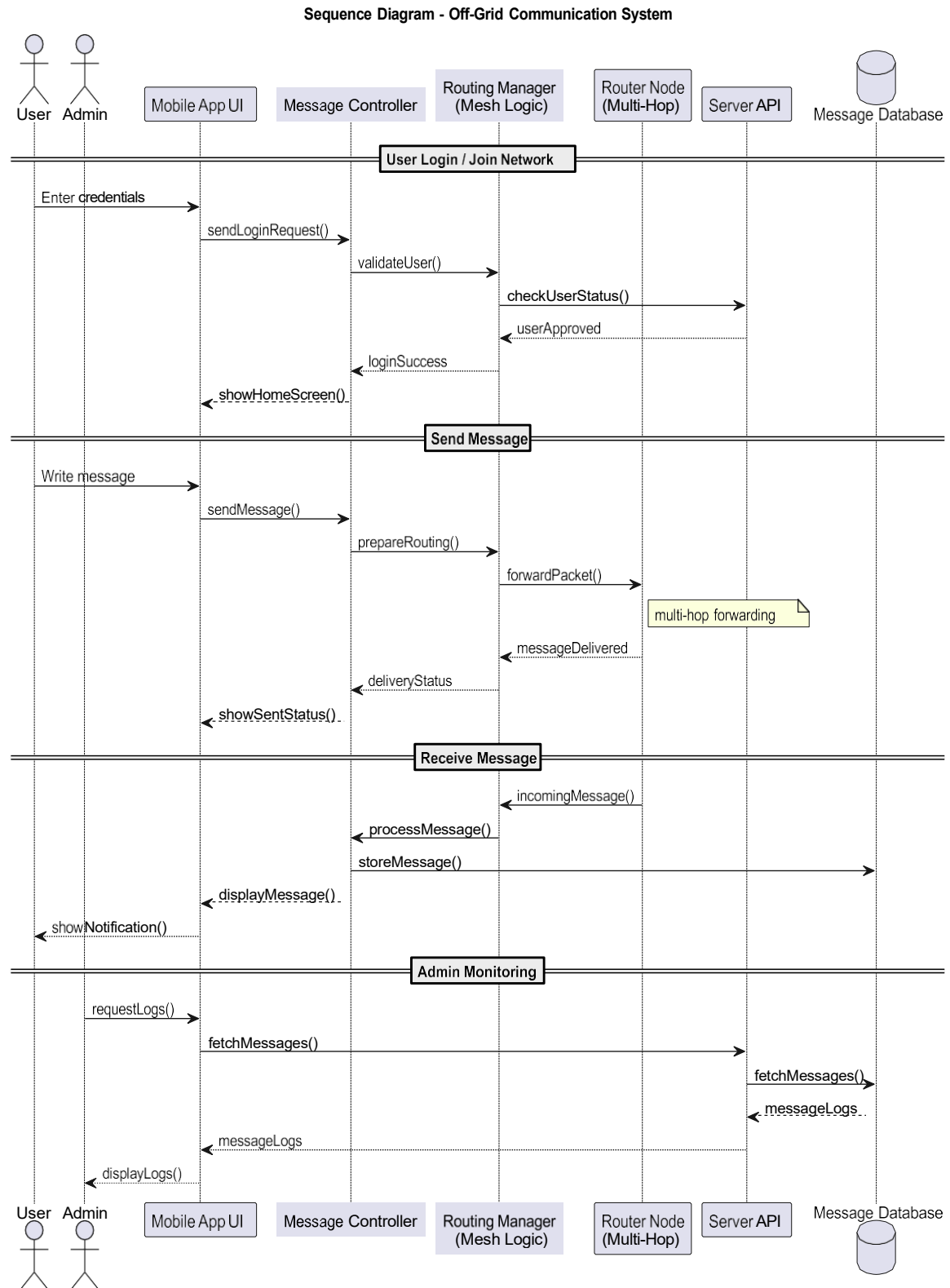


Figure 4.6: Sequence Diagram

4.8 Design Class Diagram

A DCD is a part of UML and a structural diagram, which explain static structure of system and illustrates the classes, their properties, and methods and their relationships. DCDs are an integral part of the software designing process by reflecting the system's organizational structure and the relationships between different components.

Design class diagrams are essential for the offered Off-Grid Communication System. This is attributed to the inherent structural complexity that arises from the inclusion of the mobile app, the Linux server, the routing mechanism, and the administrative control module. The aforementioned components integrate to support offline communication, user message management, and system supervision.

Figure 4.7 provides an illustration of the system's principal components with class names such as User, Admin, Message, Media, Group, Linux Server, Router, and Log Manager. Each of the components is tasked with a specific function, such as message management, logging, data authentication, and data routing. The modularity of the interconnections represented in this framework design enhances scalability and ease of management of the system.

Design Class Diagram - Off-Grid Communication System

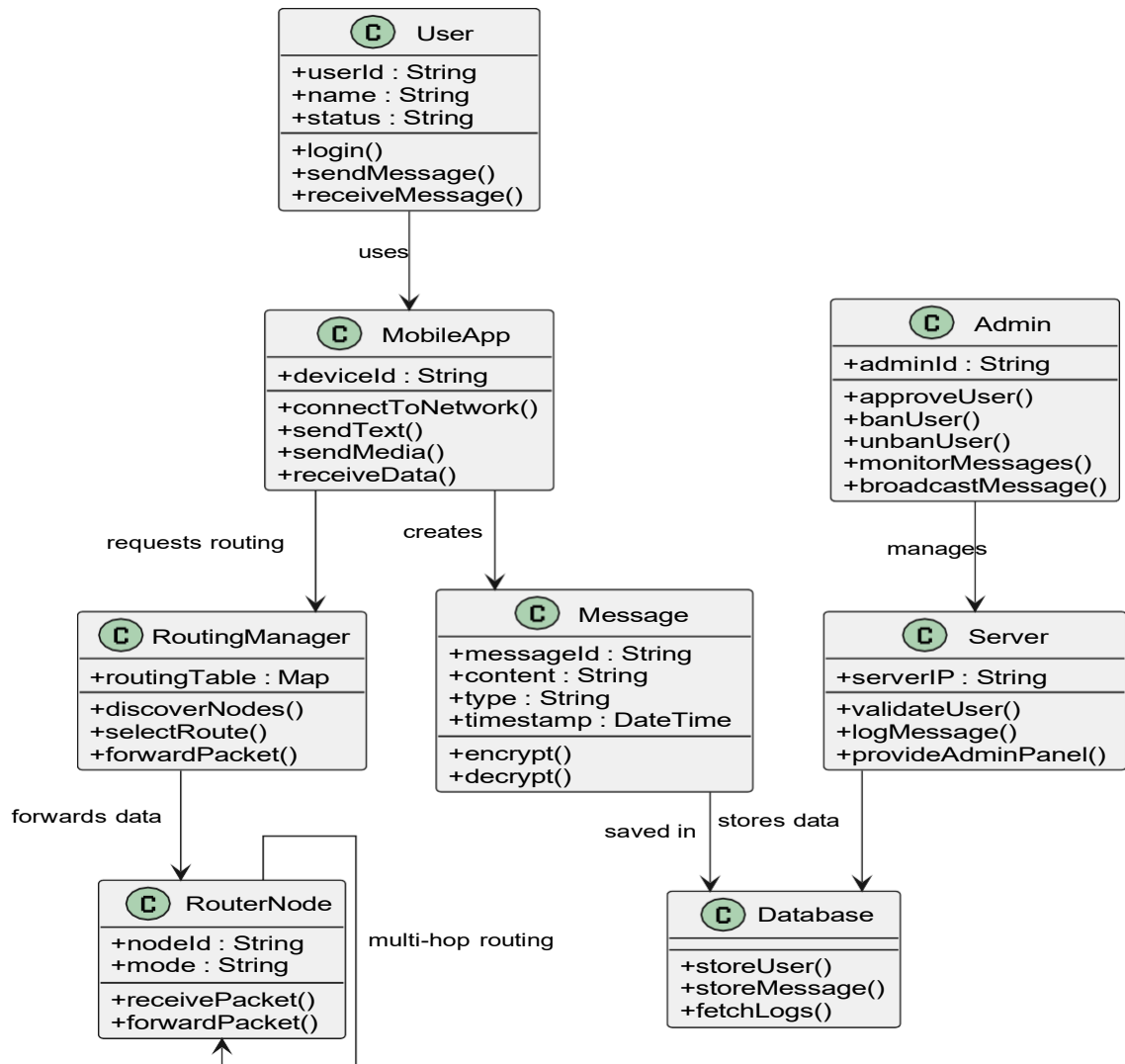


Figure 4.7: Class Diagram

4.9 Architectural Diagrams

Designed architectural diagrams holistically represent the organization of a system and the interrelation of its components. These diagrams lay the groundwork for the system and facilitate the organization of system components. They assist system designers and stakeholders in comprehending the system's organization, its components, functions, and the interrelation of system components. The complexity of a system is effectively managed with architectural diagrams to ascertain the organization of system components and their integration.

The proposed Off-Grid Communication System uses a Client-Server Distributed Architecture. This architecture was chosen because it allows for centralized control through a Linux server and makes it possible for numerous mobile clients that send and receive messages and media.

Three different architectural diagrams are created for representing the system architecture correctly; the Interface Design Diagram, the Component Design Diagram, and the Deployment Diagram. All these diagrams are designed to depict how a user will interact with the system, the system's construction, and how the system will be assembled in the environment.

4.9.1 Interface Design Diagram

User interaction with the system through the various GUIs is captured in the interface design diagram. This diagram captures user engagement, and provides little concern for system operation. From this diagram, the system is illustrated to be a mobile app for the users and desktop software for the management.

Interface Design Diagram - Off-Grid Communication System

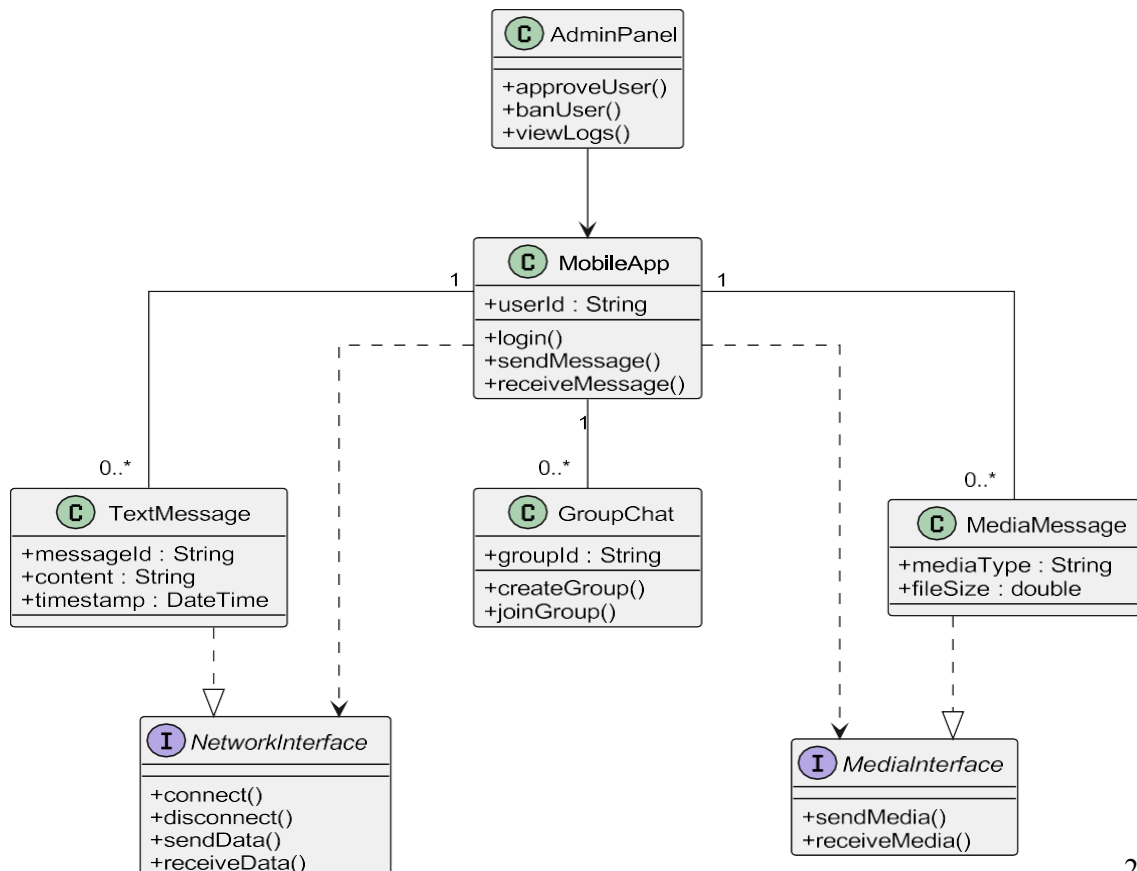


Figure 4.8: Interface Design Diagram

4.9.2 Component Level Design Diagram

The component design diagram is designed to represent software components organized by logical construction and their relationships and dependencies. This diagram captures the system's decomposition being designed into its components, and describes how these components interoperate. This diagram is captured due to the constructed system being designed to be a coalition of software components.

Component Level Design - Off-Grid Communication System

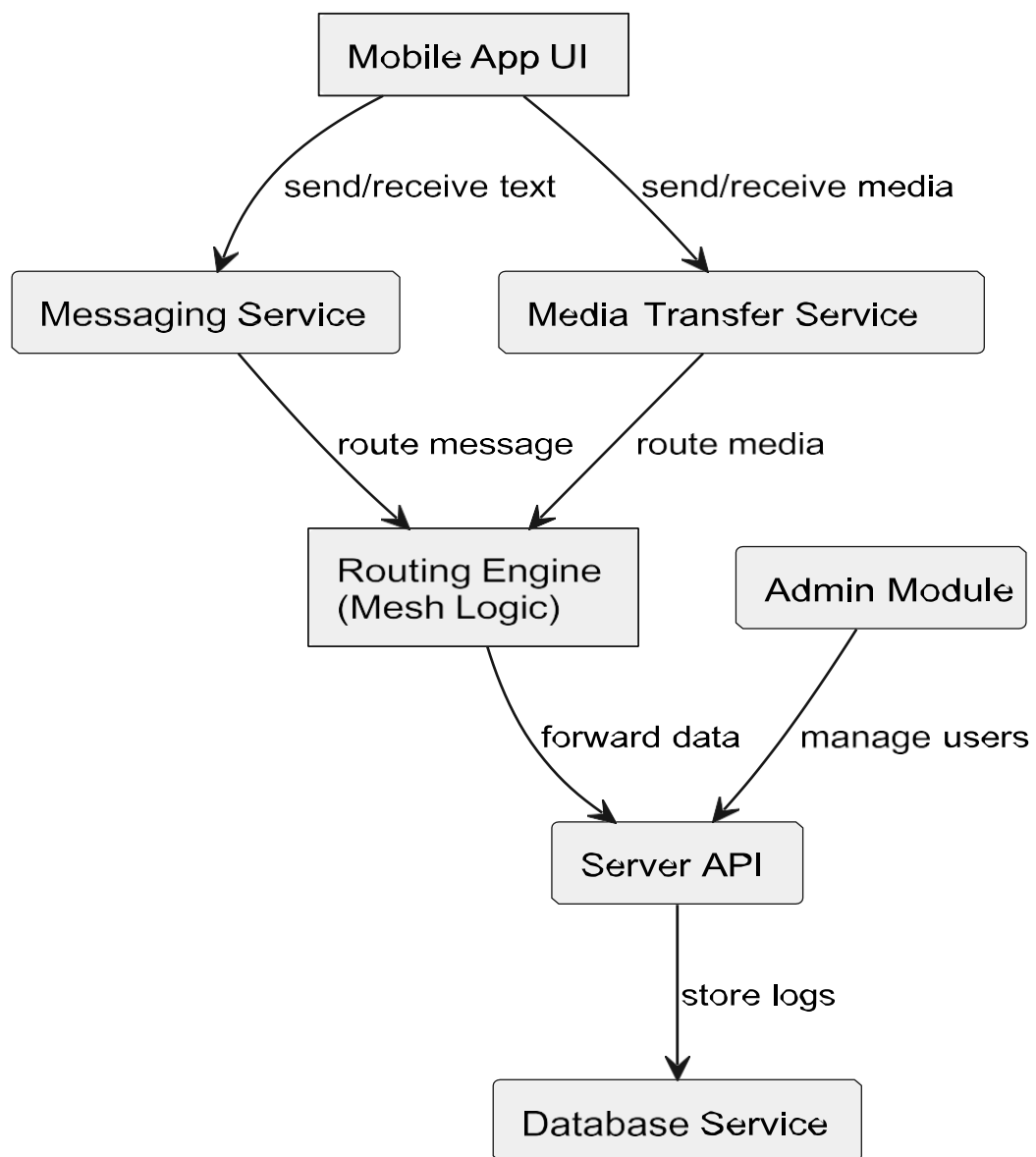


Figure 4.9: Component Level Design

4.9.3 Deployment Diagram

The deployment diagram describes how the system will be structured in the environment and how the software and hardware components will be placed. This diagram is crucial since it illustrates that the constructed system will be a coalition of software components and will be composed of hardware components such as servers, routers and mobile devices.

Deployment Diagram - Off-Grid Communication System

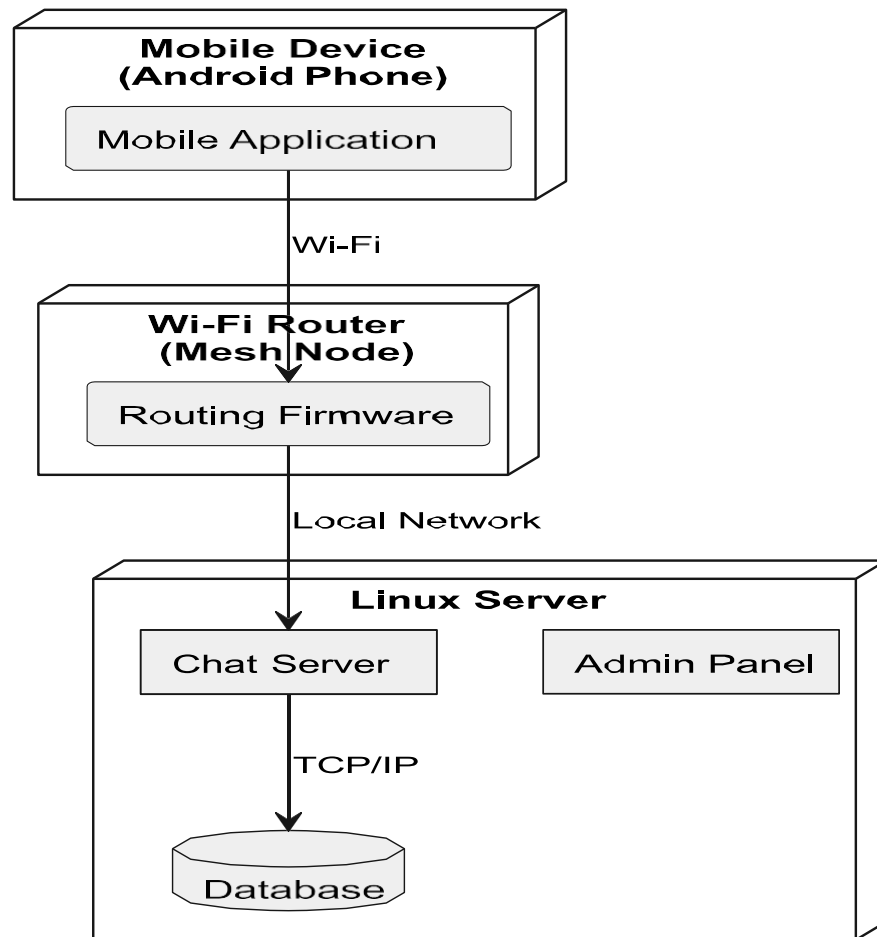


Figure 4.10: Deployment Diagram

Chapter 5

SYSTEM TESTING AND VALIDATION

In this chapter we look for testing and then check the Off-Grid Communication System that has already been set up. The goal of this chapter is to make sure that project meets user need both its functional and non-functional and works well while not connected to the internet. This chapter explains how testing works, what methods were used, how test cases were developed, and how non-functional requirements were checked. There are four main parts to the chapter: an overview of system testing, testing methods, test cases, and checking that non-functional criteria are met.

In Section 5.1, we talk about why system testing is important and give an overview of the tests that were done. Part 5.2 talks about the methods used to evaluate the system. Section 5.3 describes the test cases that were built for the system's most important functions. Section 5.4 eventually rating the system's non-functional requirements using measurable criteria.

5.1 System Testing

System testing is the process of checking a software system to make sure it meets all of its requirements and works in a variety of situations. Testing is a necessary part of making software since it finds bugs, makes sure the software is reliable, and checks that the system works before it is put into use. Testing the system thoroughly improves its quality and lowers the risk of it failing in real life.

This project included system testing to make sure that offline communication, user management, multimedia transmission, group chat, and administrative control functions all worked. We used different methods for testing, such as unit testing and integration testing. The test results show that the project meets its goals by giving strong and reliable communication even when there is no internet access. All of the important functionalities worked as expected, and the system was able to handle a local network environment.

5.2 Testing Techniques

Testing techniques are systematic perspective used for evaluate the quality and correctness of software. Different testing methods are utilized based on how complicated the system

is, how risky it is, and how well it works. This project used Unit Testing, and Integration Testing to confirm important parts were correct and worked as expected.

5.2.1 Unit Testing

Unit testing check that all component are functions correctly by testing each module individually. In this project, unit testing was conducted to test if modules such as managing messages, registering users, and performing admin tasks work as intended. Each module and function was executed separately, and the results showed that all first-stage modules and functions performed the required actions without any faults.

5.2.2 Integration Testing

Testing each module of the project and the modules of a mobile client, server, admin panel, and network modules was accomplished through integration testing. In integration testing, real participation in sending and receiving multimedia files and messages in real-time control via different devices was used to determine that the modules worked successfully.

5.3 Test Cases

With regard to mitigating risks, the use of test cases ensures that the most important goals have been achieved. Three test cases that addressed the most important functions of this system, e.g messaging, user registration, and system management, have been created. These were the most prioritized ones. These cases were defined in a framework of critical criteria.

5.3.1 Test Case 1: User Registration and Approval

This test case checks that the user registration and administrative approval process works.

The goal of this test is to make sure that only people who are allowed to can get into the system. Table 5.4 shows all the information about the test scenario.

Table 5.4: Test Case 1

GENERAL INFORMATION			
Test Stage:	Unit System Regression	Functionality Interface Acceptance	Integration Performance Pilot
Test Date:	28/12/2025	System Date	27/12/2025
Tester:	Project Team	Test Case Number:	TC-01
Testcase Description:	Verify user registration and admin approval functionality		
Results:	Pass		
Introduction			
Requirements to be tested:	User registration and admin approval		
Roles and Responsibilities:	Admin: Approve or reject user User: Submit registration request		
Set Up Procedures:	Start server and admin panel Connect mobile device to local network		
Stop Procedures:	Disconnect user and stop server		
Environmental Needs			
Hardware:	Linux server, Android smartphone, wireless router		
Software:	Python server, admin panel, mobile client		
Procedural Requirement:	Server, Admin and all Mobile devices connected with same local network and test should be perform without internet access.		

TEST	
Test Items and Features:	User registration Admin approval
Input Specifications:	User ID submission
Procedural Steps:	1. User submits registration request 2. Admin approves user 3. User gains access
Expected Results of Case:	Only approved users can access the system
Actual Results	
Output Specifications:	User appears in connected device list Test result: Passed

5.3.2 Test Case 2: Text and Multimedia Messaging

In that test case verifies real-time text and multimedia communication between users. It ensures reliable offline data transmission.

Results: Pass

Key outcome: Successful delivery of text, image, and file messages without internet connectivity.

5.3.3 Test Case 3: Admin Monitoring and Control

By this test case validates admin capabilities such as banning/unbanning users and monitoring messages.

Results: Pass

Key outcome: Admin actions reflected instantly across the system.

5.4 Non-Functional Requirements Validation

Its specify the quality attributes of a system, and these requirements certify the performance of a system smoothly, reliably, and is easy to use.

Table 5.5: List of Common Non-Functional Requirements

Property	Measures
Speed	Measure: Message delivery time Result: Messages delivered within seconds in local network
Size	Measure: Application size Result: Lightweight mobile and desktop applications
Ease of Use	Measure: Training time Result: New users learned system usage within two attempts
Reliability	Measure: System uptime Result: Stable operation during offline testing
Robustness	Measure: Recovery after failure Result: System recovered successfully after restart
Portability	Measure: Supported platforms Result: Server side Runs on Linux and Mobile application Runs on Android

Chapter 6

CONCLUSION

This chapter gives us the project's overall thoughts and about its problems and possible ways to make it better in the future. This chapter's goal is to summarize the project's overall work, highlight the main achievements, and assess how well the project's goals were realized. There are two parts to this chapter. Section 6.1 talks about the project's conclusion, and Section 6.2 talks about the system's problems and gives more ideas for future projects.

6.1 Conclusion

The project we successfully built and planned a Off-Grid Communication System that lets people talk to each other without needing an internet connection. The technology was made to fix the problems with present communication solutions that don't work well when there is no internet or in an emergency. The main features of the proposed system include real-time text messaging, multimedia sharing, group chat, and a single administrative interface for user control and monitoring. Combining a Linux-based server with long-range wireless routers improved coverage and made communication more dependable within a small local network.

The developed system met its main goals by providing a communication solution that works, is stable, and can be used in offline situations. System testing confirmed dependable message delivery, effective administrative oversight, and successful multimedia transmission without internet connectivity. The results show that the proposed system is feasible for tasks such as responding to disasters, communicating in remote areas, maintaining safety on campuses, and managing emergencies. It innovates the way resilient communication methods connect the internet, while still maintaining the way the system provides communication without the internet.

6.2 Limitations and Future Work

However, when proposed method viable, it has several constraints. It relies on using the user's local appliances and, therefore, user-configured networks, which makes it less feasible for bigger deployments and the way it currently employs security. Particularly in the case of end-to-end encryption and the most complex authentication deployments, the system performs various functions based on the user's network connectivity, hardware, and routing infrastructure. First of all, to make data more secure and private, the system performs multiple functions based on the user's network connectivity, hardware, and routing infrastructure. Complex authentication can make larger deployments easier. Additionally, in all instances, the system's performance the same as in larger, complex emergencies. End-to-end encryption mixed in with the right emergency cryptography and authentication, together with complex authentication and easily user-configurable networks, will improve the system in general.

